16. Causal Discovery

A consolidated, mathematical lecture companion

English notes based on the lecture slides *Causal Discovery* (Prof. Sanghack Lee, GSDS SNU; partly from lectures by Elias Bareinboim, Daniel Kumor and Ilya Shpitser)

The central question is how much of the causal diagram can be recovered from data alone, under which assumptions, and with which algorithms.

These notes consolidate the slides and cover constraint-based learning (faithfulness, skeletons, v-structures, Markov equivalence, the PC algorithm with Meek's rules, and the conservative, order-independent and FCI extensions), score-based learning (likelihood, BIC, GES, NOTEARS, DAGMA), parametric approaches (additive noise models, LiNGAM via ICA, score matching), and the appendices on GES operators, post-nonlinear models, Bayesian scores and CI tests. Physical PDF page numbers are listed in Section E.

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1 The structure-learning problem

Purpose. State what causal discovery takes as input and returns as output, and the assumptions needed.

Earlier lectures assumed the diagram $\mathcal{G}$ induced by an unknown SCM was given. Causal discovery goes the other way: from $P(\mathbf{V})$ (or samples) to $\mathcal{G}$. A cartoon: does the rooster's crow cause the sun to rise? A table of binary observations of (rooster, sun) shows them almost always equal, but the data alone do not say which way the arrow goes.

Structure learning. *Input:* a data set assumed sampled i.i.d. from $P(\mathbf{V})$. *Output:* a set of graphs consistent with what we know about the data (hopefully including $\mathcal{G}$). It is an unsupervised learning problem. Extensions: multiple data sets, observational and interventional data, unknown interventions, ...

1.1 DAG models

We focus on DAG models: a causal structure on $\mathbf{V}$ is a DAG with one node per variable, each directed edge representing a direct functional relationship. DAG models correspond to Markovian models, i.e. *causal sufficiency*: no unmeasured confounder of the measured variables. Three equivalent definitions, illustrated on $A \to B$, $A \to C$, $B \to D$, $C \to D$, $D \to E$:

- **Factorisation:** $P(A, B, C, D, E) = P(E \mid D)P(D \mid B, C)P(C \mid A)P(B \mid A)P(A)$.
- **Local Markov property:** $(C \perp\!\!\!\perp B \mid A)$, $(D \perp\!\!\!\perp A \mid B, C)$, $(E \perp\!\!\!\perp A, B, C \mid D)$.
- **Global Markov property:** if $\mathbf{A}$ is d-separated from $\mathbf{B}$ by $\mathbf{C}$ in $\mathcal{G}$, then $\mathbf{A} \perp\!\!\!\perp \mathbf{B} \mid \mathbf{C}$ in $P(\mathbf{V})$.

1.2 Faithfulness

The global Markov property is one-directional: every d-separation gives a CI, equivalently every dependence gives a d-connection. A complete graph is compatible with every $P(\mathbf{V})$, so we need minimality, an assumption that reverses the implication.

Definition 1.1 (Faithfulness)**.** For all $\mathbf{A}, \mathbf{B}, \mathbf{C}$: $\mathbf{A}$ is d-separated from $\mathbf{B}$ given $\mathbf{C}$ in $\mathcal{G}$ *iff* $\mathbf{A} \perp\!\!\!\perp \mathbf{B} \mid \mathbf{C}$ in $P(\mathbf{V})$.

Faithfulness is an extra property that a $P(\mathbf{V})$ factorising according to $\mathcal{G}$ may satisfy. Most such distributions are faithful (unfaithful ones have measure zero). However: in finite samples unfaithful and nearly unfaithful distributions are hard to tell apart, and there are many more of the latter; nature does not pick distributions at random and may evolve unfaithful configurations (homeostasis, equilibria). In practice one should justify why faithfulness is sensible for the problem.

Example 1.2 (Violation by cancellation)**.** $X = \epsilon_X$, $Z = X + \epsilon_Z$, $Y = X - Z + \epsilon_Y$ (edges $X \xrightarrow{1} Z$, $Z \xrightarrow{-1} Y$, $X \xrightarrow{1} Y$). X and Y are not d-separated by $\emptyset$, yet $Y = X - (X + \epsilon_Z) + \epsilon_Y = -\epsilon_Z + \epsilon_Y$, so $X \perp\!\!\!\perp Y$: the two paths cancel exactly.

2 Constraint-based structure learning

Purpose. Find the CIs that hold, rule out inconsistent graphs, and return what is left.

2.1 What constraints a DAG encodes

Example: $X \to W \to Y$, $X \to Z \leftarrow Y$. Of the candidate statements

$$(1)\ X \perp\!\!\!\perp Y \mid W, \quad (2)\ X \perp\!\!\!\perp Y, \quad (3)\ X \perp\!\!\!\perp Y \mid W, Z, \quad (4)\ X \perp\!\!\!\perp Y \mid Z,$$

only (1) is forced by the graph; (2)–(4) are dependences ($X \to W \to Y$ open; the collider Z opened). The data must exhibit the graph's independences for the graph to be valid.

With faithfulness, d-separation in $\mathcal{G} \Leftrightarrow$ CI in P, so CIs refine the possible DAGs. Without any CI information the best guess is the *complete undirected* graph (every pair adjacent, no orientation).

Adjacency. If for $A, B \in \mathbf{V}$ some $\mathbf{S} \subseteq \mathbf{V} \setminus \{A, B\}$ gives $A \perp\!\!\!\perp B \mid \mathbf{S}$, then A and B are not adjacent. (Under faithfulness, adjacent variables are dependent given every set.) In the example the only removed edge is X–Y (separated by $\{W\}$); the result, called the *skeleton*, is $X - W - Y$, $X - Z - Y$.

Orientation. For non-adjacent A, B there is a separator $\mathbf{S}$. If $A - C - B$ (an *unshielded triple*) and $C \notin \mathbf{S}$, orient $A \to C \leftarrow B$, an *unshielded collider* (v-structure); otherwise C is an unshielded non-collider (e.g. $A \to C \to B$). In the example the separator $\{W\}$ of X, Y does not include Z, so $X \to Z \leftarrow Y$; this rules out $X \to Z \to Y$, $X \leftarrow Z \leftarrow Y$ and $X \leftarrow Z \to Y$. Can anything else be oriented? $W \in \mathbf{S}$ makes $X - W - Y$ a non-collider, but its direction is undetermined.

2.2 Markov equivalence

The three DAGs $X \to W \to Y$, $X \leftarrow W \leftarrow Y$, $X \leftarrow W \to Y$ (each with $X \to Z \leftarrow Y$) entail the same independences: they are *observationally* (Markov) *equivalent*; the local Markov property gives the same CIs. A *Markov equivalence class* (MEC) is a set of Markov-equivalent DAGs.

Theorem 2.1 (Verma and Pearl 1990)**.** *Two DAGs represent the same statistical model iff they have the same skeleton and the same unshielded colliders.*

All DAGs in a class are represented by one mixed graph (directed and undirected edges), the *pattern* or CPDAG (completed partially directed acyclic graph). For three variables (Eberhardt 2017), the 25 DAGs fall into 11 classes; e.g. the chain, reversed chain and fork on X, Y, Z form one class, while the collider $X \to Y \leftarrow Z$ is alone in its class.

2.3 The PC algorithm

Basic idea: (1) non-adjacent vertices have a separator; (2) for an unshielded collider $X \to Z \leftarrow Y$, the separator must not contain Z.

Algorithm 1 PC algorithm (Spirtes and Glymour)

start with the complete undirected graph $\mathcal{G}$ on n vertices
for $i = 0, \ldots, n-2$ **do** ▷ skeleton
  for every adjacent pair A, B in $\mathcal{G}$ **do**
    if $\exists \mathbf{S}$ with $|\mathbf{S}| = i$, $\mathbf{S} \subseteq \mathrm{adj}_{\mathcal{G}}(A) \cup \mathrm{adj}_{\mathcal{G}}(B) \setminus \{A, B\}$, and $A \perp\!\!\!\perp B \mid \mathbf{S}$ **then**
      remove $A - B$; sepset$[\{A, B\}] \leftarrow \mathbf{S}$
for every non-adjacent A, B with $A - C - B$ **do** ▷ v-structures
  if $C \notin$ sepset$[\{A, B\}]$ **then** orient $A \to C \leftarrow B$
orient as many remaining edges as possible without creating new v-structures or directed cycles (Meek's rules)

Meek's rules.

1. Orient $b - c$ as $b \to c$ if there is $a \to b$ with a, c non-adjacent (else a new v-structure $a \to b \leftarrow c$ would appear).
2. Orient $a - b$ as $a \to b$ if there is a chain $a \to c \to b$ (else a cycle).
3. Orient $a - b$ as $a \to b$ if there are two chains $a - c \to b$ and $a - d \to b$ with c, d non-adjacent (if $b \to a$, Rule 2 would force $c \to a$ and $d \to a$, a new v-structure at a).
4. Orient $a - b$ as $a \to b$ if there are two chains $a - c \to d$ and $c \to d \to b$ with c, b non-adjacent and a, d adjacent.

Example 2.2 (Walk-through)**.** True (unknown) DAG: $A \to B$, $A \to C$, $B \to D$, $C \to D$. CIs revealed by the tests: $B \perp\!\!\!\perp C \mid A$ (fork blocked by A, collider D closed) and $A \perp\!\!\!\perp D \mid \{B, C\}$ (both paths blocked).

- $|\mathbf{S}| = 0$: all pairs dependent; nothing removed.
- $|\mathbf{S}| = 1$: $B \perp\!\!\!\perp C \mid A$: remove $B - C$, sepset$[\{B, C\}] = \{A\}$; other tests dependent.
- $|\mathbf{S}| = 2$: $A \perp\!\!\!\perp D \mid \{B, C\}$: remove $A - D$, sepset$[\{A, D\}] = \{B, C\}$. Skeleton: $A - B$, $A - C$, $B - D$, $C - D$.
- V-structures: $B - D - C$ with $D \notin \{A\}$, so $B \to D \leftarrow C$; $B - A - C$ with $A \in \{A\}$, not a collider; $A - B - D$, $A - C - D$ with $B, C \in \{B, C\}$, not colliders.
- Meek's rules: the slides apply Rule 1 to $A - B$ (using $B \to D$ with A, D non-adjacent) and to $A - C$, obtaining $A \to B$, $A \to C$, $B \to D \leftarrow C$, which matches the true DAG.

A caution on the last step: Rule 1 requires an arrow *into* B from a vertex non-adjacent to A, whereas $B \to D$ points out of B. Strictly, no Meek rule fires here, and the CPDAG is $A - B$, $A - C$, $B \to D \leftarrow C$. Its class contains $A \to B, A \to C$; $B \to A \to C$; $C \to A \to B$ (each with $B \to D \leftarrow C$), while $B \to A \leftarrow C$ is excluded because it would add a v-structure. The true DAG is one member of this class.

Exercise. Apply PC to $A \to B$, $A \to C$, $B \to D$, $C \to D$, $D \to E$ (the earlier five-variable DAG). The v-structure $B \to D \leftarrow C$ is found, and Rule 1 orients $D \to E$ (since $B \to D$ with B, E non-adjacent).

2.4 Extensions

Conservative PC (Ramsey, Spirtes and Zhang 2006). Split faithfulness into *adjacency faithfulness* (adjacent ⇒ no separator) and *orientation faithfulness*: if $X \to Z \leftarrow Y$ is an unshielded collider, X, Y are dependent given every subset of $\mathbf{V} \setminus \{X, Y\}$ containing Z; otherwise every separator contains Z. Algorithmically, check all subsets of X's and of Y's potential parents: if Z is in no separating set, orient a v-structure; if in all, a non-collider; otherwise mark the triple "unfaithful" (shown underlined in the output). This makes the algorithm robust to violations of orientation faithfulness at modest extra cost.

Order-independent PC (Colombo and Maathuis 2014). PC removes an edge as soon as a separator is found; a false-negative CI test early on changes later adjacency sets, so the output depends on the variable ordering (the slides show skeletons varying across 25 random orderings). Remedies: average over random orderings, or remove edges only after finishing all tests with the same conditioning-set size; conservative orientation is too restrictive, so a majority rule over separating sets is introduced.

Fast Causal Inference (FCI) relaxes causal sufficiency (allows latent confounders). Example: true graph $X \to Y$, $W \to Z$, and a latent U with $U \to Y$, $U \to Z$. After CI-based edge removal

the skeleton is $X - Y - Z - W$; FCI outputs $X \circ\!\!\rightarrow Y \leftrightarrow Z \leftarrow\!\!\circ W$, indicating an unmeasured confounder of Y and Z. PAG edge marks:

$X_1 \quad X_2$ (no edge)	X_1 and X_2 are not adjacent
$X_1 \circ\!\!\rightarrow X_2$	X_2 is not an ancestor of X_1
$X_1 \circ\!\!-\!\!\circ X_2$	no set d-separates X_1 and X_2
$X_1 \rightarrow X_2$	X_1 is a cause of X_2
$X_1 \leftrightarrow X_2$	there is a latent common cause of X_1 and X_2

2.5 Summary

CI constraints yield partial information about the structure. With a CI oracle (all tests correct), PC is complete (it recovers the full CPDAG). CI testing in practice uses partial correlation (linear Gaussian), Fisher's exact or χ^2 tests (categorical), kernel-based or model-based tests (Appendix D). Relaxing sufficiency and/or faithfulness is active research.

3 Score-based approaches

Purpose. Search for the graph that best fits the data with a complexity penalty.

3.1 Why scores?

Constraint-based methods rely on CI tests, which can be unreliable in finite samples; a single wrong test can propagate errors through skeleton and orientation. Score-based methods define a function measuring how well a graph fits the data (and how simple it is) and search over graphs. The paradigms are complementary; hybrid methods (e.g. MMHC) combine them.

Why it is possible. Relative to the true structure (e.g. $E \rightarrow A \leftarrow B$, $A \rightarrow S$), *adding* an arc increases the number of parameters and encodes wrong causal assumptions; *missing* an arc cannot be compensated by fitting parameters accurately and also misses causal structure.

Intuition. True graph $A \rightarrow B \leftarrow C$, proposed $A \rightarrow B \rightarrow C$. In truth $A \perp\!\!\!\perp C$ but $A \not\perp\!\!\!\perp C \mid B$; the proposed graph says $A \not\perp\!\!\!\perp C$ and $A \perp\!\!\!\perp C \mid B$. Data will likely show $A \perp\!\!\!\perp C$ and $A \not\perp\!\!\!\perp C \mid B$, so the proposal fits poorly. Fit is assessed by the (log-)likelihood.

3.2 Log-likelihood for DAG models

With $P(\mathbf{V}) = \prod_i P(V_i \mid Pa_{\mathcal{G}}(V_i))$ and models $P_{\theta_i}(V_i \mid Pa_{\mathcal{G}}(V_i))$, the log-likelihood of data $D = \{\mathbf{v}^{(j)}\}_{j=1}^{n}$ is

$$\ell(\theta; D) = \sum_{j=1}^{n} \sum_{i=1}^{k} \log P_{\theta_i}(v_i^{(j)} \mid \mathbf{pa}_{\mathcal{G}}^{(j)}(V_i)),$$

a *decomposable* likelihood: each θ_i is fitted separately on the part of the data involving $P(V_i \mid Pa_{\mathcal{G}}(V_i))$.

For the true $A \rightarrow B \leftarrow C$ and proposed $B \rightarrow A$, $C \rightarrow B$:

$$\ell_{\text{left}} = \textstyle\sum_j \left[\log P_A(a_j) + \log P_B(b_j \mid a_j, c_j) + \log P_C(c_j)\right],$$
$$\ell_{\text{right}} = \textstyle\sum_j \left[\log P_A(a_j \mid b_j) + \log P_B(b_j \mid c_j) + \log P_C(c_j)\right].$$

Maximised, the true graph has the higher likelihood: with saturated local models, $\ell_{\text{left}}/n - \ell_{\text{right}}/n = I(A; C \mid B) > 0$ (Appendix C).

The problem. Compare the true graph with $B \to A$, $C \to A$, $C \to B$ (complete): $\ell_{\text{right}} = \sum_j [\log P_A(a_j \mid b_j, c_j) + \log P_B(b_j \mid c_j) + \log P_C(c_j)]$ fits *better*, because the left model $\{P : A \perp\!\!\!\perp C\}$ is a submodel of the right (all of $P(A, B, C)$). With likelihood alone the best model is always a complete graph: overfitting. Too many parameters fit well but miss the sparse structure; reward fit, but punish parameters.

3.3 BIC

$$\text{BIC} = -2\,\ell(\hat{\theta}; D) + m \log n,$$

with m the number of parameters; lower is better. For a linear Gaussian network $X_i = \beta_{0i} + \sum_{X_k \in Pa(X_i)} \beta_{ki} X_k + \epsilon_i$, $\epsilon_i \sim \mathcal{N}(0, \sigma_i^2)$, node i contributes $|Pa(X_i)|+2$ parameters (slopes, intercept, variance), so $m = \sum_i (|Pa(X_i)| + 2) = |E| + 2|V|$. BIC is a regularisation over graph structures.

3.4 Search and GES

Naively listing all DAGs and scoring each is impossible (super-exponentially many DAGs, possibly intractable likelihoods). Local search moves between states (DAGs) by local moves able to reach any DAG, ideally finding the best state in a polynomial sequence of moves. **Greedy Equivalence Search** (GES; Chickering 2002) defines states and moves on equivalence classes:

1. Start with the empty graph.
2. *Addition phase:* repeatedly move to the best class obtained by adding a single edge, until no improvement. (Adding an edge to a class can lead to different classes depending on the representative: adding $B - C$ to $\{A \to B \;\; C,\; A \leftarrow B \;\; C\}$ leads either to $\{A \to B \leftarrow C\}$ or to the class of chains/forks on A, B, C.)
3. *Removal phase:* repeatedly move to the best class obtained by removing a single edge, until no improvement.
4. Return the current class.

Theorem 3.1 (Chickering 2002)**.** *Under faithfulness and as $n \to \infty$, GES identifies the true equivalence class.*

Operators (on CPDAGs). Insert$(u, v, \mathbf{T})$: add $u \to v$ for non-adjacent u, v, with $\mathbf{T} \subseteq \text{Ne}(v) \setminus \text{Adj}(u)$ (undirected neighbours of v not adjacent to u); orient every $t \in \mathbf{T}$ as $t \to v$ (possibly creating unshielded colliders); convert to a CPDAG. Delete$(u, v, \mathbf{H})$: remove the edge between u and v, with $\mathbf{H} \subseteq \text{Ne}(v) \cap \text{Adj}(u)$; orient $h \to v$ and $h \to u$ for each $h \in \mathbf{H}$; convert to a CPDAG. Each operator yields a valid CPDAG, so the search stays in class space; at each step GES picks the operator (with its $u, v, \mathbf{T}$ or $\mathbf{H}$) with the best score improvement.

Example 3.2. True DAG $A \to B \leftarrow C$ (a v-structure, a unique class). Forward: start empty; Insert$(A, B, \emptyset)$ gives $A - B$ (undirected, as $A \to B$ and $A \leftarrow B$ are equivalent); Insert$(C, B, \{A\})$ (valid since $A \in \text{Ne}(B) \setminus \text{Adj}(C)$) gives $A \to B \leftarrow C$; no further insertion helps. Backward: no deletion helps. Output $A \to B \leftarrow C$.

3.5 Continuous optimisation: NOTEARS and DAGMA

NOTEARS (Zheng et al. 2018) learns a linear SEM (noise not necessarily Gaussian) by black-box optimisation. Example: $X = \epsilon_X$, $Z = aX + \epsilon_Z$, $Y = bX + cZ + \epsilon_Y$, i.e.

$$\begin{pmatrix} X \\ Z \\ Y \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ a & 0 & 0 \\ b & c & 0 \end{pmatrix} \begin{pmatrix} X \\ Z \\ Y \end{pmatrix} + \begin{pmatrix} \epsilon_X \\ \epsilon_Z \\ \epsilon_Y \end{pmatrix} \iff \begin{pmatrix} X \\ Z \\ Y \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b + ac & c & 1 \end{pmatrix} \begin{pmatrix} \epsilon_X \\ \epsilon_Z \\ \epsilon_Y \end{pmatrix}.$$

With data $\mathbf{X} \in \mathbb{R}^{n\times d}$ and coefficients $W \in \mathbb{R}^{d\times d}$,

$$\min_{W\in\mathbb{R}^{d\times d}} \frac{1}{2n}\|\mathbf{X} - \mathbf{X}W\|_F^2 + \lambda\|W\|_1 \quad \text{subject to} \quad h(W) = \operatorname{tr}\left(e^{W\circ W}\right) - d = 0.$$

Why this characterises acyclicity: for a binary adjacency matrix B, B^k counts walks of length k ($B^0 = I$), $e^B = \sum_k B^k/k!$, and $\operatorname{tr}(e^B)$ counts closed walks weighted by $1/k!$, including the d trivial ones; hence $\operatorname{tr}(e^B) = d$ iff B is a DAG. $\operatorname{tr}(e^{W\circ W})$ counts weighted cycles (including 0-length ones) and is smooth, so the problem is solved by an augmented Lagrangian (NOTEARS: Non-combinatorial Optimization via Trace Exponential and Augmented lagRangian for Structure learning).

NOTEARS-MLP (Zheng et al. 2020) learns general mechanisms f: $\min_f L(f)$ subject to $\mathcal{G}(f) \in$ DAGs, with $L(f) = \frac{1}{n}\sum_{j=1}^d \ell(x_j, f_j(\mathbf{X}))$, $\mathbb{E}[X_j \mid X_{\mathrm{pa}(j)}] = g_j(f_j(X))$, $\mathbb{E}f_j(X) = 0$, where $k \to j$ is an edge iff f_j depends on X_k. A surrogate for W: f_j is independent of X_k iff $\|\partial_k f_j\|_{L^2} = 0$, so set $[W(f)]_{kj} = \|\partial_k f_j\|_{L^2}$ and solve $\min_{f_j\in H^1(\mathbb{R}^d)} L(f)$ s.t. $h(W(f)) = 0$ (H^1: Sobolev space of square-integrable functions with square-integrable derivatives). With MLPs $\mathrm{MLP}(u; A^{(1)}, \ldots, A^{(h)}) = \sigma(A^{(h)}\cdots\sigma(A^{(2)}\sigma(A^{(1)}u)))$, the output is independent of u_k if the k-th column of $A^{(1)}$ is zero, so the problem becomes

$$\min_\theta \frac{1}{n}\sum_{j=1}^d \ell(x_j, \mathrm{MLP}(\mathbf{X};\theta_j)) + \lambda\|A_j^{(1)}\|_{1,1} \quad \text{s.t.} \quad h(W(\theta)) = 0,$$

with

$$[W(\theta)]_{kj} = \|k\text{-th column of } A_j^{(1)}\|_2.$$

DAGMA (Bello, Aragam and Ravikumar 2022). The trace-exponential penalty is smooth but numerically demanding (matrix exponentials) and its optimisation can be difficult for larger graphs. DAGMA uses a log-determinant penalty

$$h_{\mathrm{DAGMA}}(W) = -\log\det(sI - W\circ W) + d\log s, \qquad s > 0 \text{ (e.g. } s = 1).$$

Acyclic graphs get zero penalty, graphs with cycles positive penalty, which grows quickly near highly cyclic regions and gives a useful signal before the graph is acyclic. It keeps the NOTEARS setup, optimising fit, sparsity and acyclicity together, $\min_\theta Q(f_\theta; \mathbf{X}) + \beta_1\|\theta\|_1 + h_{\mathrm{DAGMA}}(W(\theta))$, for linear or nonlinear SEMs, and reports faster optimisation with competitive accuracy. It is not a new causal assumption, only another optimisation device for enforcing DAGness.

3.6 Summaries

Score-based. BIC and GES guarantees are large-sample; with finite data correctness is not guaranteed. Local search is intractable if the score cannot be evaluated in polynomial time. If the local models $P(V_i \mid Pa_{\mathcal{G}}(V_i))$ are misspecified, the graph is likely wrong. No hidden variables are assumed.

Constraint- vs. score-based. Constraint-based (PC, FCI): CI tests prune edges, orientation via sepsets and Meek's rules; output a CPDAG (PAG for FCI); sensitive to CI errors. Score-based (GES, NOTEARS, DAGMA): a score is optimised over graph space; tends to do better with small samples, requires correct model specification. Both assume faithfulness and (except FCI) causal sufficiency; hybrids (MMHC) combine both.

4 Parametric approaches

Purpose. Use functional-form assumptions to orient edges that are indistinguishable from CIs alone.

Under what conditions can we orient $X \to Y$ versus $Y \to X$? With two variables there are no CIs to exploit; asymmetries in the joint distribution must be used.

Additive noise model (ANM). $y = f_y(x) + u_y$ with $x \perp\!\!\!\perp u_y$. If the distribution is known to come from an ANM, compare

$$\text{H1: } y = f_y(x) + u_y,\ x \perp\!\!\!\perp u_y\ (x \to y) \qquad \text{vs.} \qquad \text{H2: } x = f_x(y) + u_x,\ y \perp\!\!\!\perp u_x\ (y \to x).$$

Two main cases: (1) f non-linear; (2) f linear but u_y non-Gaussian.

4.1 Non-linear functions

For non-linear f (generically), an ANM holds in only one direction: in the wrong direction the residual's spread depends on the regressor. Procedure, given (x_i, y_i):

1. Fit both directions with any ML method: $\hat{y} = \hat{f}_y(x)$ and $\hat{x} = \hat{f}_x(y)$.
2. Compute residuals (guesses of u): $\hat{u}_x = x - \hat{f}_x(y)$, $\hat{u}_y = y - \hat{f}_y(x)$.
3. Test residual independence (e.g. HSIC): $\hat{u}_y \perp\!\!\!\perp x$ suggests $X \to Y$; $\hat{u}_x \perp\!\!\!\perp y$ suggests $Y \to X$. If neither holds, the ANM assumption is suspect.

4.2 Linear functions with non-Gaussian noise: LiNGAM

Cramér's decomposition theorem: if the sum of two independent real random variables is Gaussian, both summands are Gaussian; by induction the same holds for any finite sum. This complements the CLT: sums approach Gaussianity but cannot be exactly Gaussian unless every summand is.

LiNGAM (linear non-Gaussian acyclic model). For $y = ax + u_y$ with non-Gaussian errors,

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ a & 1 \end{pmatrix} \begin{pmatrix} u_x \\ u_y \end{pmatrix}, \qquad x = u_x, \quad y = au_x + u_y.$$

In the backward regression of x on y, the residual is a mixture of u_x and u_y that is uncorrelated with, but (by Darmois–Skitovich/Cramér-type results) not independent of, y; in the forward direction the residual u_y is independent of x.

The trick: ICA. Suppose independent non-Gaussian sources $s_1, \ldots, s_n$ are observed only through a mixture $\mathbf{x} = A\mathbf{s}$. How can A and $\mathbf{s}$ be recovered? By the CLT intuition, $x = as_1 + bs_2$ looks more Gaussian than s_1 or s_2; ICA finds directions maximising non-Gaussianity. Measures:

- *Kurtosis* (fourth cumulant) $\kappa(y) = \frac{\mathbb{E}[(y-\mu)^4]}{\mathrm{Var}(y)^2} - 3$: zero for Gaussians, positive for super-Gaussian (heavy-tailed, e.g. Laplace), negative for sub-Gaussian (light-tailed, e.g. uniform); simple but outlier-sensitive.
- *Negentropy* $J(y) = H(y_{\text{Gauss}}) - H(y) \geq 0$, with y_{Gauss} Gaussian with the same mean and variance; $J(y) = 0$ iff y is Gaussian (the Gaussian maximises entropy for a given variance).

ICA takes $\mathbf{x} = A\mathbf{s}$ and returns an unmixing matrix W_{ICA} with $\hat{\mathbf{s}} = W_{\text{ICA}}\mathbf{x}$, up to permutation and scaling (including sign): among all $\mathbf{w}^\top\mathbf{x}$, the maximally non-Gaussian ones are the sources (Hyvärinen and Oja 2000). In the two-variable case, run ICA on (x, y) and look for the zero in the mixing matrix.

LiNGAM (Shimizu et al. 2006). For ordered variables $x_i = \sum_{j<i} b_{ij}x_j + e_i$, i.e. $\mathbf{x} = B\mathbf{x} + \mathbf{e}$, $(I - B)\mathbf{x} = \mathbf{e}$, $\mathbf{x} = (I - B)^{-1}\mathbf{e}$. Let $W = I - B$. ICA gives $W_{\text{ICA}} = PDW$ (P a permutation, D diagonal: the permutation, scaling and sign indeterminacies), but the LiNGAM structure resolves them:

1. $\mathbf{e} = (I - B)\mathbf{x} = W\mathbf{x}$, with W a row-permuted, rescaled version of $I - B$.
2. Run ICA: $W_{\text{ICA}} = \text{ICA}(\mathbf{X})$.
3. Permute the rows of W_{ICA} so that the diagonal has no zeros (the unique such permutation, since $I - B$ has a non-zero diagonal and is permutable to lower-triangular), then rescale rows so the diagonal is one: $W = \text{normalize}(\text{rowpermute}(W_{\text{ICA}}))$.
4. $\hat{B} = I - W$; a causal order is the one making $\hat{B}$ lower-triangular.

Summary. Given an ANM $y = f(x) + u_Y$ with f non-linear or U_Y non-Gaussian, the direction of the arrow is identifiable, assuming no latent confounding ($X \perp\!\!\!\perp U_Y$).

4.3 Score matching (NoGAM)

(Montagna et al. 2023.) Consider an additive non-linear model $X_i = f_i(\text{PA}_i) + N_i$ with arbitrary noise. The *score function* is the gradient of the log-density, $s(X) = \nabla_X \log p(X)$, $s_i(X) = \partial_{X_i} \log p(X)$ (unrelated to BIC/GES scores): it describes how the log-density changes when one variable is moved. Score matching estimates $s(X)$ from samples without estimating $p(X)$, and the score identifies *leaf* nodes; recursively removing leaves gives a topological order, after which extra edges are pruned.

Why leaves are special. Writing $\log p(X) = \sum_k \log p_k(N_k)$ with $N_k = X_k - f_k(\text{PA}_k)$,

$$s_l(X) = \partial_{N_l} \log p_l(N_l) \quad \text{for a leaf } X_l,$$

$$s_i(X) = \partial_{N_i} \log p_i(N_i) - \sum_{k \in \text{CH}_i} \partial_{N_k} \log p_k(N_k)\, \partial_{X_i} f_k \quad \text{otherwise.}$$

For a leaf, regressing X_l on the rest gives residual $R_l = X_l - q(X_{-l}) = N_l$ (in population), so $s_l(X)$ is a function of R_l alone; for non-leaves the children terms remain and R_i alone generally cannot predict $s_i(X)$. Population statement (under ANM identifiability conditions):

$$X_i \text{ is a leaf} \iff \min_h \mathbb{E}\big[\{h(R_i) - s_i(X)\}^2\big] = 0.$$

Algorithm (NoGAM). (1) Estimate $\hat{s}(X) \approx \nabla_X \log p(X)$. (2) For each remaining node, regress X_i on X_{-i} and compute residuals $\hat{R}_i$. (3) Predict $\hat{s}_i(X)$ from $\hat{R}_i$ and record the MSE. (4) Declare the node with smallest MSE a leaf, remove it, and repeat. (5) Reverse the removal order to get a topological order, then prune extra edges. This recovers an order without assuming Gaussian noise.

A GES operators in detail

Insert$(u, v, \mathbf{T})$ adds $u \to v$ (with $u \notin \text{Adj}(v)$), orients $t - v$ as $t \to v$ for every $t \in \mathbf{T}$ ($t \notin \text{Adj}(u)$), and completes the PDAG to a CPDAG. Key set: $\mathbf{T} \subseteq \text{Ne}(v) \setminus \text{Adj}(u)$, and $(\text{Ne}(v) \cap \text{Adj}(u)) \cup \mathbf{T}$ must be a clique; full validity also requires every semi-directed path from v to u to pass through this blocking set. In the v-structure example, after $A - B$ the correct move is $\text{Insert}(C, B, \{A\})$.

Delete$(u, v, \mathbf{H})$ removes the edge between u and v ($u - v$ or $u \to v$), orients $h - v$ as $h \to v$ and $h - u$ as $h \to u$ for every $h \in \mathbf{H}$ (h points into both endpoints), and completes to a CPDAG. Key set: $\mathbf{H} \subseteq \text{Ne}(v) \cap \text{Adj}(u)$, and $(\text{Ne}(v) \cap \text{Adj}(u)) \setminus \mathbf{H}$ must be a clique. These are Chickering's (2002) operators; XGES reparametrises the same score-relevant sets (Nazaret et al. 2025, "Extremely greedy equivalence search").

B Post-nonlinear causal model (optional)

PNL extends the ANM with an invertible transformation of the effect: for $X_1 \to X_2$, $X_2 = g_2\{f_1(X_1) + E_2\}$, $E_2 \perp\!\!\!\perp X_1$, g_2 invertible. Then $g_2^{-1}(X_2) = f_1(X_1) + E_2$, i.e. $E_2 = g_2^{-1}(X_2) - f_1(X_1)$. Find ψ_2, ϕ_1 such that $E_2 = \psi_2(X_2) - \phi_1(X_1)$ is independent of X_1; if this works in one direction but not the other, orient the edge. Identifiability holds generically, with special functional/noise cases still ambiguous (Zhang and Hyvärinen 2009).

C Scores in detail

Why the left likelihood is larger. For left $A \to B \leftarrow C$ and right $B \to A$, $C \to B$, with saturated local models the maximised population log-likelihoods are negative conditional entropies:

$$\ell_{\text{left}}/n = -H(A) - H(C) - H(B \mid A, C), \qquad \ell_{\text{right}}/n = -H(C) - H(B \mid C) - H(A \mid B).$$

If the truth is the collider, $A \perp\!\!\!\perp C$ and $H(A) + H(C) + H(B \mid A, C) = H(A, B, C)$, so

$$\ell_{\text{left}}/n - \ell_{\text{right}}/n = H(A \mid B) + H(B \mid C) - H(A) - H(B \mid A, C) = I(A; C \mid B).$$

Conditioning A on B helps the right graph, but dropping A from B's parents hurts more, exactly when $A \not\perp\!\!\!\perp C \mid B$.

Bayesian Dirichlet scores (categorical data). The BD score (Cooper and Herskovits) computes the probability of a network given data, under assumptions. Drawbacks: the user must specify α_{ijk} for every variable–parent configuration, and equivalent structures need not get equal scores. BDe is the score-equivalent variant obtained via an equivalent sample size α; BDeu sets $\alpha_{ijk} = \alpha/(r_i q_i)$, spreading α uniformly over local CPT cells (not a uniform prior over graphs). With i indexing variables, j parent configurations (q_i of them) and k values (r_i of them), and the Dirichlet as conjugate prior of the multinomial,

$$\text{BD}(\mathcal{G}, D; \alpha) = \prod_{i=1}^{N} \text{BD}(X_i \mid Pa_{X_i}; \alpha_i) = \prod_{i=1}^{N} \prod_{j=1}^{q_i} \frac{\Gamma(\alpha_{ij})}{\Gamma(\alpha_{ij} + n_{ij})} \prod_{k=1}^{r_i} \frac{\Gamma(\alpha_{ijk} + n_{ijk})}{\Gamma(\alpha_{ijk})},$$

α_i being the imaginary (equivalent) sample size. $\alpha_{ijk} = 1$ gives the K2 score; $\alpha_{ijk} = 1/2$ the BD score with Jeffreys' prior; $\alpha_{ijk} = \alpha/(r_i q_i)$ the BDeu score (with $\alpha_i = \alpha$ for all X_i).

BIC (maximisation form). Dividing by -2,

$$\text{BIC}(\mathcal{G}, D) = \sum_{i=1}^{N} \Big[\ell_i(\hat{\theta}_i) - \frac{\log n}{2} q_i(r_i - 1)\Big], \qquad \ell_i(\hat{\theta}_i) = \sum_{m=1}^{n} \log \hat{P}_i(x_i^{(m)} \mid \pi_i^{(m)}),$$

which is score-equivalent. BIC is the large-sample Laplace approximation of log BDeu; they agree up to $O(1)$ as $n \to \infty$.

D CI tests for causal discovery

χ^2 test (categorical). Tests $P(X, Y \mid Z) = P(X \mid Z)P(Y \mid Z)$ for all z. Example (2008 General Social Survey, "Where do you get most of your information about current news events?"; credit Yibi Huang):

Age	TV	Internet	Newspapers	Other	Total
18–29	109	92	25	36	262
30–49	272	157	88	63	580
50+	345	59	165	63	632
Total	726	308	278	162	1474

(Did the way people get news change with age?) For each level of Z, expected counts under CI are $E(x, y \mid z) = N(x, z)N(y, z)/N(z)$, and

$$\chi^2(Z = z) = \sum_{x,y} \frac{(O(x, y \mid z) - E(x, y \mid z))^2}{E(x, y \mid z)}, \qquad \chi^2_{\text{total}} = \sum_z \chi^2(Z = z),$$

with $df = (|X| - 1)(|Y| - 1)|Z|$; the p-value compares χ^2_{total} with the χ^2_{df} distribution.

Partial correlation (linear).

$$r_{XY \cdot Z} = \frac{r_{XY} - r_{XZ} r_{YZ}}{\sqrt{1 - r^2_{XZ}}\sqrt{1 - r^2_{YZ}}}$$

for a single conditioning variable (for multivariate Z use recursion or regression residuals). If it remains high, X and Y are correlated beyond Z; if it drops, Z explains most of their correlation.

HSIC (non-linear marginal independence). Tests $H_0 : X \perp\!\!\!\perp Y$ without assuming linearity. With kernels k, l, Gram matrices K, L and $H = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$, $\widehat{\text{HSIC}}(X, Y) = \frac{1}{n^2}\operatorname{tr}(KHLH)$. With characteristic kernels, population HSIC is zero iff $X \perp\!\!\!\perp Y$. Null calibration by permutation, gamma approximation or asymptotics (Gretton et al. 2005). Useful when marginal dependence may be non-linear and non-Gaussian; for conditional independence use KCI.

Kernel conditional independence test (KCI). Targets $H_0 : X \perp\!\!\!\perp Y \mid Z$ via RKHS conditional covariances: represent X, Y, Z by kernels K_X, K_Y, K_Z; regress out Z in the RKHS, e.g. $R_Z = \epsilon(K_Z + \epsilon I)^{-1}$, $\tilde{K}_X = R_Z K_X R_Z$, $\tilde{K}_Y = R_Z K_Y R_Z$; test whether the residualised dependence $T_{\text{KCI}} = \frac{1}{n}\operatorname{tr}(\tilde{K}_X \tilde{K}_Y)$ is zero. It replaces Fisher-z or χ^2 tests in PC-style discovery when non-linear CI matters, at higher computational cost and with sensitivity to kernel bandwidth and regularisation (Zhang, Peters, Janzing and Schölkopf 2011).

E Source coverage

The source deck has 165 physical pages (103 numbered slides with build-ups).

Topic	Physical pages
Credits, problem, DAG models, faithfulness	1–15
Constraints, skeleton, orientation, Markov equivalence	16–43
PC algorithm, Meek's rules, walk-through	44–59
Conservative PC, order independence, FCI, summary	60–72
Score-based: likelihood, BIC, GES, NOTEARS, DAGMA, summaries	73–107
Parametric: ANM, non-linear, LiNGAM/ICA, score matching	108–146
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